

Problem Overview:

The logarithm of the price of an agricultural product A, is influenced by local environmental and climatic factors, with the observational model over a period of 3 months given by:

$$y_t = \mu + 0.5x_t + \epsilon_t \quad (1)$$

where:

$y_t \Rightarrow$ log-price of A on day t for $t = 1, \dots, 90$

$x_t \Rightarrow$ environmental condition on day t

$\mu \Rightarrow$ unknown baseline price level with a prior $\rightarrow \pi(\mu) \sim N(0,100)$

$\epsilon \Rightarrow$ iid $N(0,1)$ random variables

The overall environmental condition is evaluated using daily rainfall measurements along with the condition of the previous day, given by the following model:

$$x_t = R_t\beta + 0.6x_{t-1} + \eta_t \quad (2)$$

where:

$t = 1, 2, \dots, 90$

$x_0 \equiv 12$

$\beta \Rightarrow$ an unknown constant with a prior $\rightarrow \pi(\beta) \sim N(0,100)$

$\eta_t \Rightarrow$ iid $N(0,1)$ random variables

Let $\mathcal{D} = (R_1, R_2, \dots, R_n, y_1, y_2, \dots, y_n)$ with $n = 90$ be the collection of all observations.

1. Let $X_t = (x_t, \mu, \beta)$. Rewriting models (1) and (2) into a state space model (SSM) involving X_t and y_t for $t > 1$.

The SSM consists of two models:

$Y_k = A_k X_k + \xi_k \rightarrow$ Measurement Equation Model (MEM)

$X_k = F_k X_{k-1} + \epsilon_k \rightarrow$ State Equation Model (SEM)

Therefore, in the given problem the MEM and SEM equations will be rewritten as follows:

MEM:

$Y_t = A_t X_t + \xi_t$ where $\xi_t \sim N(0,1) \rightarrow R = 1$

$$\Rightarrow Y_t = [0.5 \ 1 \ 0] \begin{bmatrix} x_t \\ \mu \\ \beta \end{bmatrix} + \xi_t$$

where:

$$A_t = [0.5 \ 1 \ 0]$$

$$X_t = \begin{bmatrix} x_t \\ \mu \\ \beta \end{bmatrix}$$

SEM:

$X_t = F_t X_{t-1} + \epsilon_t$ where $\epsilon_t \sim N(0,1) \rightarrow Q = 1$

$$\Rightarrow X_t = \begin{bmatrix} x_t \\ \mu \\ \beta \end{bmatrix} = \begin{bmatrix} 0.6 & 0 & R_t \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_{t-1} \\ \mu \\ \beta \end{bmatrix} + \begin{bmatrix} \eta_t \\ 0 \\ 0 \end{bmatrix}$$

$$\text{where: } F_t = \begin{bmatrix} 0.6 & 0 & R_t \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad X_{t-1} = \begin{bmatrix} x_{t-1} \\ \mu \\ \beta \end{bmatrix}$$

$$\epsilon_t = \begin{bmatrix} \eta_t \\ 0 \\ 0 \end{bmatrix}$$

2a. Kalman filtering is the appropriate filtering technique which provides recursive estimates of forecast and analysis distributions for X_t through the forecasting of the mean and covariance of X_t at each iteration. The forecasted mean and covariance may then be used for evaluation of the predictive distribution function of $y_{t+1}|y_{1:t}$.

2b. Based on the MEM and SEM equations obtained in question 1, define (at $t = 0$), the mean vector of X_0 to be as $\bar{x}^{a,0}$ and the covariance matrix of X_0 to be as $P^{a,0}$, therefore:

$$X_0 \sim N_d(\mu_0 \equiv \bar{x}^{a,0}, \Sigma_0 \equiv P^{a,0}) \Rightarrow X_0 \sim N_3\left(\begin{pmatrix} 12 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 100 & 0 \\ 0 & 0 & 100 \end{pmatrix}\right)$$

Therefore, for every $t = 1, 2, \dots, n$, the analysis mean and covariance matrix of X_t are denoted as $\bar{x}^{a,t}$ and $P^{a,t}$ respectively. Furthermore, the forecast mean and covariance matrix will be denoted as $\bar{x}^{f,t}$ and $P^{f,t}$ respectively.

Hence, for $t = 1$, the forecasted probability density function of X_1 is obtained from the SEM:

Since $X_1 = F_1 X_0 + \epsilon_t$ is the marginal normal distribution of X_1 , the mean vector $\bar{x}^{f,1}$ and the covariance matrix $P^{f,1}$ are as follows:

$$\bar{x}^{f,1} = F_1 \bar{x}^{a,0} = \begin{bmatrix} 0.6 & 0 & R_t \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \\ 0 \end{bmatrix}$$

$$P^{f,1} = F_1 P^{a,0} F_1^T + Q_1 = \begin{bmatrix} 0.6 & 0 & R_t \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 100 & 0 \\ 0 & 0 & 100 \end{bmatrix} \begin{bmatrix} 0.6 & 0 & 0 \\ 0 & 1 & 0 \\ R_t & 0 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore$ The mean and the covariance matrix of the forecast state distribution are obtained below:

$$\begin{aligned} \bar{x}^{f,2} &= F_2 \bar{x}^{a,1} = F_2 \left(F_1 \begin{pmatrix} 12 \\ 0 \\ 0 \end{pmatrix} \right) & P^{f,2} &= F_2 P^{a,1} F_2^T + Q_2 \\ & & &= F_2 (F_1 P^{a,0} F_1^T + Q_1) F_2^T + Q_2 \\ \bar{x}^{f,3} &= F_3 \bar{x}^{a,2} = F_3 \left(F_2 \left(F_1 \begin{pmatrix} 12 \\ 0 \\ 0 \end{pmatrix} \right) \right) & P^{f,3} &= F_3 P^{a,2} F_3^T + Q_3 \\ & & &= F_3 (F_2 (F_1 P^{a,0} F_1^T + Q_1) F_2^T + Q_2) F_3^T + Q_3 \\ \Rightarrow \bar{x}^{f,t} &= F^t \bar{x}^{a,0} & \Rightarrow P^{f,t} &= F_t P^{a,t-1} F_t^T + Q_t \\ \Rightarrow \bar{x}^{f,t} &= F_t \bar{x}^{a,t-1} \end{aligned}$$

Therefore, the forecast state distribution is given as $X_t \sim N_3(\bar{x}^{f,t}, P^{f,t})$

In order to obtain a forecast observation distribution, we assimilate Y_t using the MEM and the prior on $X_t \sim N_d(\bar{x}^{f,t}, P^{f,t})$:

$$Y_t = A_t X_t + \xi_t \text{ where } \xi_t \sim N(0, 1)$$

$$\Rightarrow Y_{t+1} = A_{t+1} X_{t+1} + \xi_{t+1} \text{ where } X_{t+1} = F_{t+1} X_t + \epsilon_{t+1}$$

$$\Rightarrow Y_{t+1} = A_{t+1} (F_{t+1} X_t + \epsilon_{t+1}) + \xi_{t+1}$$

$$\begin{aligned} \therefore E(Y_{t+1}) &= E(A_{t+1} (F_{t+1} X_t + \epsilon_{t+1}) + \xi_{t+1}) \\ &= E(A_{t+1} F_{t+1} X_t + A_{t+1} \epsilon_{t+1}) + E(\xi_{t+1}) \\ &= E(A_{t+1} F_{t+1} X_t) + E(A_{t+1} \epsilon_{t+1}) + 0 \\ &= A_{t+1} F_{t+1} E(X_t) \\ &= A_{t+1} F_{t+1} \bar{x}^{a,t} \end{aligned}$$

$\rightarrow$ Mean of the forecasted distribution

$$\begin{aligned} Cov(Y_{t+1}) &= Cov(A_{t+1} F_{t+1} X_t + A_{t+1} \epsilon_{t+1} + \xi_{t+1}) \\ &= Cov(A_{t+1} F_{t+1} X_t) + Cov(A_{t+1} \epsilon_{t+1}) + Cov(\xi_{t+1}) \\ &= A_{t+1} F_{t+1} P^{a,t} F_{t+1}^T A_{t+1}^T + A_{t+1} Q A_{t+1}^T + R \end{aligned}$$

$\rightarrow$ Covariance of the forecasted distribution

$$\therefore Y_{t+1} | Y_{1:t} \sim N_1(A_{t+1} F_{t+1} \bar{x}^{a,t}, A_{t+1} F_{t+1} P^{a,t} F_{t+1}^T A_{t+1}^T + A_{t+1} Q A_{t+1}^T + R)$$

Additionally, if we were to assimilate Y_1 , we would use the MEM and the prior on $X_1 \sim N_d(\bar{x}^{f,1}, P^{f,1})$, hence obtaining the conditional PDF of $X_1^a | Y_1 \sim N_d(\bar{x}^{a,1}, P^{a,1})$ where:

$$\begin{aligned}\bar{x}^{a,1} &= \bar{x}^{f,1} + P^{f,1} A_1^T (A_1 P^{f,1} A_1^T + R)^{-1} (y_{obs}^1 - A_1^T \bar{x}^{f,1}) \\ P^{a,1} &= P^{f,1} - P^{f,1} A_1^T (A_1 P^{f,1} A_1^T + R)^{-1} A_1 P^{f,1}\end{aligned}$$

Ultimately the general equations of the Kalman analysis are:

$$\begin{aligned}\bar{x}^{a,t} &= \bar{x}^{f,t} + P^{f,t} A_t^T (A_t P^{f,t} A_t^T + R)^{-1} (y_{obs}^t - A_t^T \bar{x}^{f,t}) \\ P^{a,t} &= P^{f,t} - P^{f,t} A_t^T (A_t P^{f,t} A_t^T + R)^{-1} A_t P^{f,t}\end{aligned}$$

where $K_t \equiv P^{f,t} A_t^T (A_t P^{f,t} A_t^T + R)^{-1}$ is the Kalman gain matrix

Once the analysis PDF at time $t = 1$ is obtained, the steps above are repeated for $t = 2, \dots, n$.

2c. Based on the question 2b, the Kalman filtering algorithm starting from $t = 1$ will be initialized using the prior of X_0 as follows:

$$X_0 \sim N_3 \left(\bar{x}^{a,0} = \begin{pmatrix} 12 \\ 0 \\ 0 \end{pmatrix}, P^{a,0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 100 & 0 \\ 0 & 0 & 100 \end{pmatrix} \right)$$

Therefore, starting at $t = 1$, we assimilate Y_1 as explained in 2b, to obtain $(\bar{x}^{a,1}, P^{a,1})$.

3a. In order to obtain the posterior distribution of $(\mu, \beta | \mathcal{D})$, the filter derived in Q2 has to be run from $t = 1$ to $t = n$, and the analysis distribution has to be obtained at $t = n$, where $n = 90$.

Therefore, the analysis mean and covariance at $t = n$ are the following:

$$X_n \sim N_3 \left(\bar{x}^{a,n} = \begin{pmatrix} x_t \\ \mu \\ \beta \end{pmatrix}, P^{a,n} = \begin{pmatrix} \sigma_{xx} & \sigma_{x\mu} & \sigma_{x\beta} \\ \sigma_{\mu x} & \sigma_{\mu\mu} & \sigma_{\mu\beta} \\ \sigma_{\beta x} & \sigma_{\beta\mu} & \sigma_{\beta\beta} \end{pmatrix} \right)$$

Hence, the posterior distribution of (μ, β) given $\mathcal{D}$, denoted by $p(\mu, \beta | \mathcal{D})$ is:

$$\mu, \beta | \mathcal{D} \sim N_2 \left(\begin{pmatrix} \bar{\mu} \\ \bar{\beta} \end{pmatrix}, \begin{pmatrix} \sigma_{\mu\mu} & \sigma_{\mu\beta} \\ \sigma_{\beta\mu} & \sigma_{\beta\beta} \end{pmatrix} \right)$$

3b. The posterior distribution $p(\theta | \mathcal{D})$ is derived through the application of linear transformation to the bivariate Normal distribution obtained in 3a, since any linear combination of normally distributed variables is normally distributed as well.

Since θ is a linear combination of μ and β , the equation of θ could be rewritten as:

$$\theta = \mu - R_{91}\beta = (1 - R_{91}) \begin{pmatrix} \mu \\ \beta \end{pmatrix}$$

where $a_1 x_1 + a_2 x_2 = a^T X_n$, $X_n = \begin{pmatrix} \mu \\ \beta \end{pmatrix}$, and $a = \begin{pmatrix} 1 \\ -R_{91} \end{pmatrix} = \begin{pmatrix} 1 \\ -10 \end{pmatrix}$

Therefore, knowing the mean and covariance of μ and β derived in 3a where,

$$E(\mu, \beta) = \begin{pmatrix} \bar{\mu} \\ \bar{\beta} \end{pmatrix} \text{ and } \Sigma = \begin{pmatrix} \sigma_{\mu\mu} & \sigma_{\mu\beta} \\ \sigma_{\mu\beta} & \sigma_{\beta\beta} \end{pmatrix}$$

We can determine the mean and variance of $(\theta | \mathcal{D})$:

$$\begin{aligned}E(\theta | \mathcal{D}) &= a^T E(\mu, \beta) & \text{Var}(\theta | \mathcal{D}) &= a^T \Sigma a \\ &= (1 \quad -10) \begin{pmatrix} \bar{\mu} \\ \bar{\beta} \end{pmatrix} & &= (1 \quad -10) \begin{pmatrix} \sigma_{\mu\mu} & \sigma_{\mu\beta} \\ \sigma_{\mu\beta} & \sigma_{\beta\beta} \end{pmatrix} \begin{pmatrix} 1 \\ -10 \end{pmatrix} \\ &= \bar{\mu} - 10\bar{\beta} & &= (\sigma_{\mu\mu} - 10\sigma_{\mu\beta} \quad \sigma_{\mu\beta} - 10\sigma_{\beta\beta}) \begin{pmatrix} 1 \\ -10 \end{pmatrix} \\ & & &= \sigma_{\mu\mu} - 20\sigma_{\mu\beta} + 10^2\sigma_{\beta\beta}\end{aligned}$$

Therefore the posterior distribution $p(\theta | \mathcal{D})$ is given as: $\theta | \mathcal{D} \sim N_1(\bar{\mu} - 10\bar{\beta}, \sigma_{\mu\mu} - 20\sigma_{\mu\beta} + 10^2\sigma_{\beta\beta})$

4a. The trajectory plots of the analysis means of X_t are displayed in Figures below.

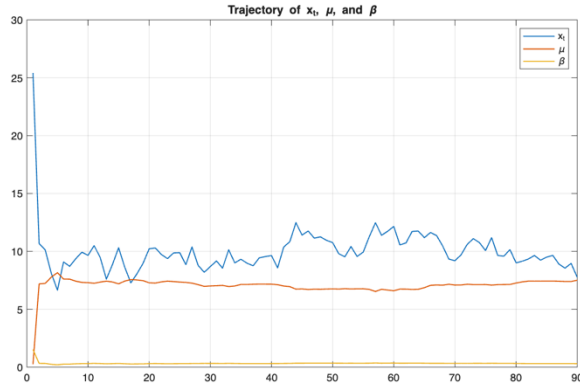


Figure 1: Trajectory Plot of the Analysis Means of X_t

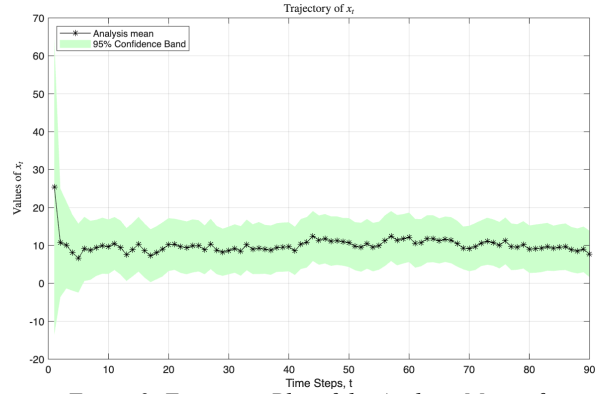


Figure 2: Trajectory Plot of the Analysis Mean of x_t

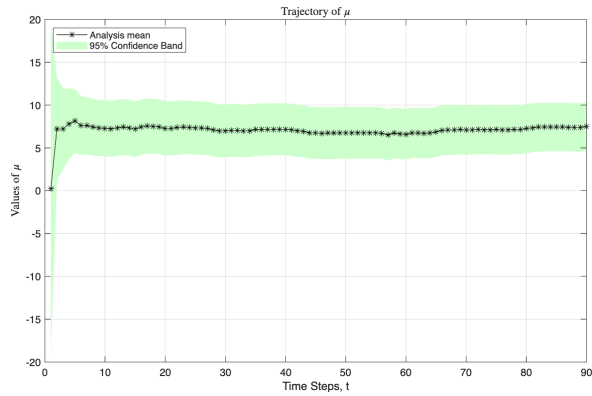


Figure 3: Trajectory Plot of the Analysis Mean of μ

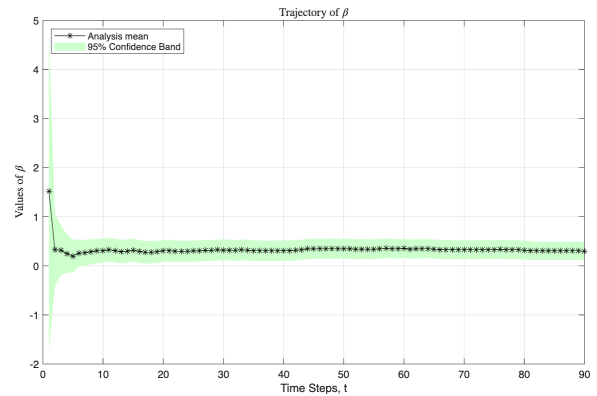


Figure 4: Trajectory Plot of the Analysis Mean of β

The confidence band in the trajectory plots above displays wide confidence bands at the start due to the priors on μ and β . The variance then stabilizes as the filter accumulates enough observations to estimate the mean parameter more precisely. Therefore, based on the plots, the Kalman filter successfully converges, since both trajectory and uncertainty stabilizes.

4b. Based on the MATLAB code provided in the Appendix, the posterior means, standard deviations and 95% credible intervals for β and μ , are the following:

According to the derivation of the posterior distribution of μ and β in 3a:

$$\begin{aligned} \mu, \beta | \mathcal{D} &\sim N_2 \left(\begin{pmatrix} \bar{\mu} \\ \bar{\beta} \end{pmatrix}, \begin{pmatrix} \sigma_{\mu\mu} & \sigma_{\mu\beta} \\ \sigma_{\beta\mu} & \sigma_{\beta\beta} \end{pmatrix} \right) \\ \Rightarrow \mu, \beta | \mathcal{D} &\sim N_2 \left(\begin{pmatrix} 7.5072 \\ 0.2975 \end{pmatrix}, \begin{pmatrix} 2.0757 & -0.1367 \\ -0.1367 & 0.0091 \end{pmatrix} \right) \\ \Rightarrow E(\mu, \beta | \mathcal{D}) &= \begin{pmatrix} 7.5072 \\ 0.2975 \end{pmatrix} \text{ and } Cov(\mu, \beta | \mathcal{D}) = \begin{pmatrix} 2.0757 & -0.1367 \\ -0.1367 & 0.0091 \end{pmatrix} \\ \Rightarrow sd(\mu, \beta | \mathcal{D}) &= \begin{pmatrix} 1.4407 \\ 0.0955 \end{pmatrix} \end{aligned}$$

95% Credible Interval for μ = [4.6834, 10.331]

95% Credible Interval for β = [0.11022, 0.48475]

4c. Figure 5 below, illustrates the 95% confidence region for (μ, β) :

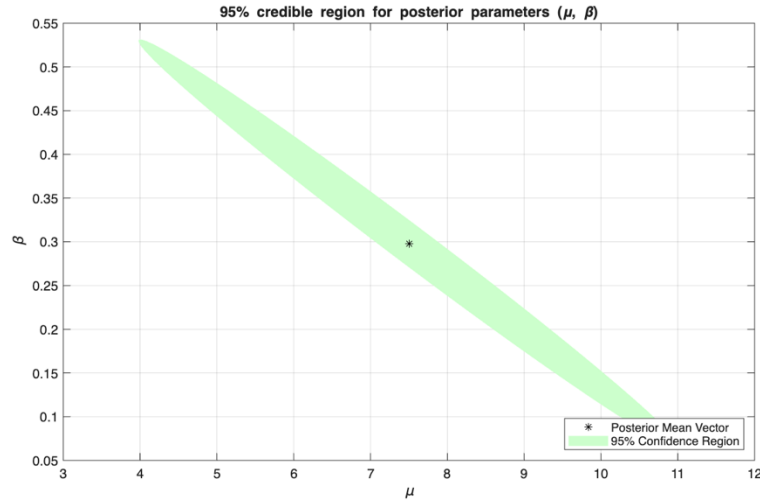


Figure 5: 95% Confidence Region for (μ, β)

Figure 5 displays a green ellipse which represents the parameters μ and β that have 95% posterior probability under the bivariate normal posterior distribution obtained through the Kalman filtering technique. The ellipse is negatively sloped, indicating a strong negative correlation between μ and β . The posterior mean is located at the center of the ellipse.

4d. Based on the MATLAB code provided in the Appendix, the posterior mean, standard deviation and the 95% credible interval for θ is presented below:

$$E(\theta|\mathcal{D}) = 4.5324$$

$$Var(\theta|\mathcal{D}) = 5.7232$$

$$sd(\theta|\mathcal{D}) = 2.3932$$

$$95\% \text{ Credible Interval for } \theta = [-0.1565, 9.2213]$$